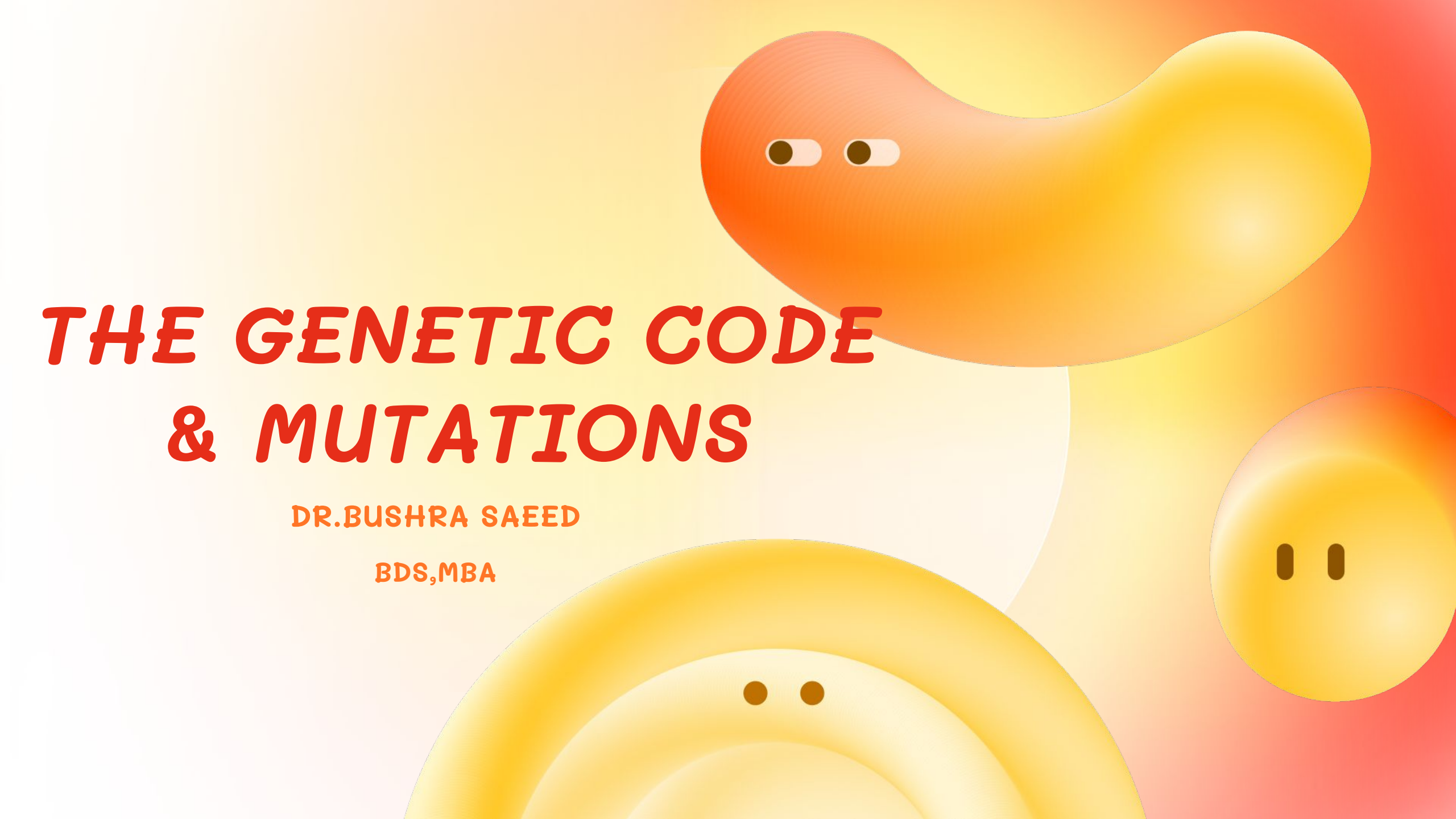


The background features a warm gradient of orange and yellow. Several large, organic, blob-like shapes in these colors are scattered across the frame. Some of these shapes contain stylized eyes, represented by two small black dots or horizontal lines. The overall aesthetic is soft and abstract.

THE GENETIC CODE & MUTATIONS

DR.BUSHRA SAEED

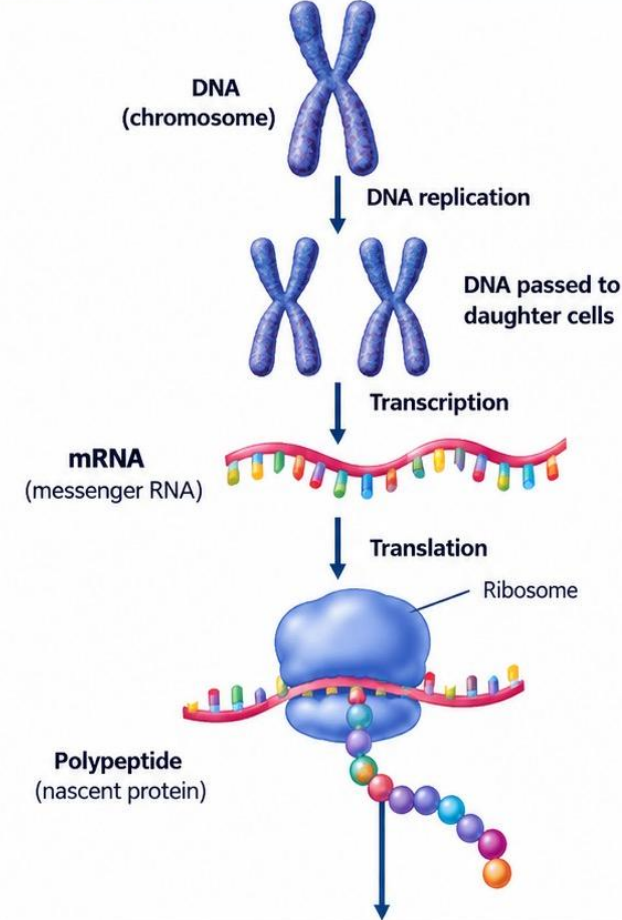
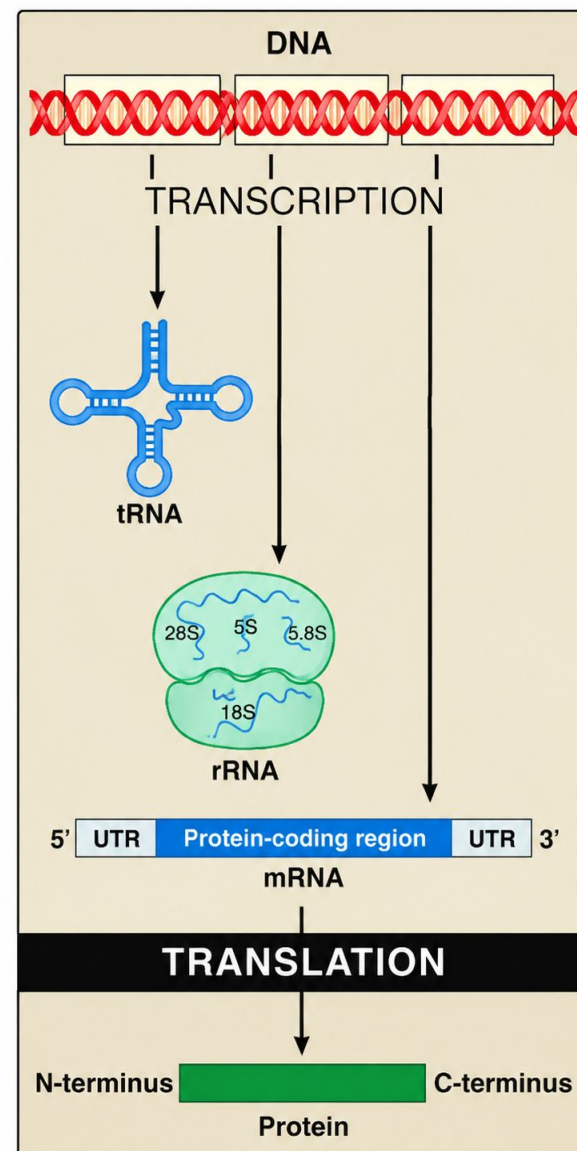
BDS,MBA

OVERVIEW — PROTEIN SYNTHESIS



- Genetic information is stored in DNA within chromosomes.
- DNA is passed to daughter cells through DNA replication.
- Gene expression occurs in two main steps:
 - ✓ **Transcription:** DNA → RNA
 - ✓ **Translation:** mRNA → Protein
- Translation is called translation because the nucleotide “language” of mRNA is converted into the amino acid “language” of a protein.
- Translation requires a genetic code that determines the correct amino acid sequence.
- A change (mutation) in the nucleotide sequence may:
 - ✓ Insert the wrong amino acid
 - ✓ Produce an abnormal protein.
 - ✓ Cause disease or, in severe cases, death.
- Newly synthesized proteins are called **nascent proteins**- undergo several processes to become functional:
 - ✓ Proper folding → gives the protein its correct shape.
 - ✓ Covalent modification → can change protein activity.
 - ✓ Targeting → directs the protein to its correct intracellular or extracellular location.

- Protein misfolding may cause:
- ✓ Protein aggregation
- ✓ Protein degradation
- ✓ Disease



DNA

- Stores genetic information.
- Replicates and is passed to daughter cells.

Transcription

- DNA → RNA

Translation

- mRNA → polypeptide (protein).
- Uses a genetic code to convert nucleotide sequence into an amino acid sequence.
- A change in the nucleotide sequence may insert the wrong amino acid, leading to an abnormal protein and possible disease or death.

Post-translational processing of nascent proteins

Folding

- Achieves correct 3D structure.

Covalent modification

- Alters activity (e.g., phosphorylation, glycosylation, etc.).

Targeting

- Directs protein to its final intra- or extracellular destination (via signal sequences).

Proteome

= the complete set of proteins expressed in a cell.



Functional protein

Misfolding can result in:

- Aggregation
- Degradation
- Disease

THE GENETIC CODE

- ❑ The genetic code is a “dictionary” that identifies the correspondence between a sequence of nucleotide bases and a sequence of amino acids.
- ❑ Each individual “word” in the code is composed of three nucleotide bases. These genetic words are called codons.

A. Codons

- ❑ Codons are presented in the mRNA language of adenine (A), guanine (G), cytosine (C), and uracil (U).
- ❑ Their nucleotide sequences are always written from the 5'-end to the 3'-end. The four nucleotide bases are used to produce the three-base codons. Therefore, 64 different combinations of bases exist, taken three at a time (a triplet code)

The Triplet Code

The genetic code is read in sets of three RNA nucleotides (codons).
Three-letter code words provide enough information to specify 20 amino acids
plus start and stop signals.

1 nucleotide

RNA nucleotides

A

Adenosine

Nc1ncnc2c1ncn2[C@@H]3O[C@H](CO)[C@@H](O)[C@H]3O

U

Uridine

O=C1NC(=O)C([C@H]2O[C@H](CO)[C@@H](O)[C@H]2O)=CN1

C

Cytidine

Nc1cc[nH]c1[C@@H]2O[C@H](CO)[C@@H](O)[C@H]2O

G

Guanosine

Nc1nc2[nH]cnc2c(=O)[nH]1[C@@H]3O[C@H](CO)[C@@H](O)[C@H]3O

4 possibilities
 $4^1 = 4$

Too few to code for 20 amino acids.

2 nucleotides

Examples of 2-nucleotide combinations

AA

AU

AC

AG

UA

UU

UC

UG

CA

CU

CC

CG

GA

GU

GC

GG

16 possibilities
 $4^2 = 16$

Still too few to code for 20 amino acids.

3 nucleotides = codon

A codon is three RNA nucleotides read together.

A

U

G

codon

Examples of codons and their meanings

A U G	→	START	start codon (start signal)
U A A	→	STOP	stop codon (stop signal)
U A G	→	STOP	stop codon (stop signal)
G G U	→	Gly	glycine (amino acid)
U U U	→	Phe	phenylalanine (amino acid)

Each codon specifies either an amino acid or a start or stop signal.

64 possibilities
 $4^3 = 64$

64 codons

Enough to encode 20 amino acids plus start/stop signals.



RNA nucleotides

A

U

C

G

Four nucleotide types in RNA.

20 common amino acids

Ala	Arg	Asn	Asp	Cys	Gln	Glu	Gly	His	Ile
Leu	Lys	Met	Phe	Pro	Ser	Thr	Trp	Tyr	Val

Proteins are built from these 20 amino acids.

64 codons in total

61

sense codons
(specify amino acids)

1

 start codon

3

 stop codons

Multiple codons can specify the same amino acid (degeneracy of the code).

The Genetic Code: Amino Acid Chart

The genetic code is a set of rules used by cells to translate mRNA codons into the amino acids that make up proteins.

KEY

- AUG start codon
- Stop codons

SECOND BASE

FIRST BASE	SECOND BASE				THIRD BASE
	U	C	A	G	
	U UUU Phe UUC Phe UUA Leu UUG Leu	UCU Ser UCC Ser UCA Ser UCG Ser	UAU Tyr UAC Tyr UAA Stop UAG Stop	UGU Cys UGC Cys UGA Stop UGG Trp	
	C CUU Leu CUC Leu CUA Leu CUG Leu	CCU Pro CCC Pro CCA Pro CCG Pro	CAU His CAC His CAA Gln CAG Gln	CGU Arg CGC Arg CGA Arg CGG Arg	
	A AUU Ile AUC Ile AUA Ile AUG Met (Start)	ACU Thr ACC Thr ACA Thr ACG Thr	AAU Asn AAC Asn AAA Lys AAG Lys	AGU Ser AGC Ser AGA Arg AGG Arg	
	G GUU Val GUC Val GUA Val GUG Val	GCU Ala GCC Ala GCA Ala GCG Ala	GAU Asp GAC Asp GAA Glu GAG Glu	GGU Gly GGC Gly GGA Gly GGG Gly	

mRNA CODON

HOW TO USE THE CHART

To identify the amino acid for an mRNA codon, read the bases in order: **first base**, then **second base**, then **third base**.

EXAMPLES

- 1 **AUG** → **methionine** (start codon)
 - First base: **A** (row A)
 - Second base: **U** (column U)
 - Third base: **G** (third position G) → Met (Start)
- 2 **CAG** → **glutamine**
 - First base: **C** (row C)
 - Second base: **A** (column A)
 - Third base: **G** (third position G) → Gln (glutamine)
- 3 **UAG** → **stop**
 - First base: **U** (row U)
 - Second base: **A** (column A)
 - Third base: **G** (third position G) → Stop



AUG start codon
codes for **methionine (Met)** and signals the start of translation.



UAA, UAG, UGA
stop codons
signal the end of translation.

There are 64 mRNA codons ($4 \times 4 \times 4$). They specify **20 amino acids** and **3 stop signals**.

1. How to translate a codon:

- This table can be used to translate any codon and, thus, to determine which amino acids are coded for by an mRNA sequence. For example, the codon AUG codes for methionine ([Met] [Note: AUG is the initiation (start) codon for translation.]

		Second base				
		U	C	A	G	
First base	U	UUU } Phenyl-alanine UUC } UUA } Leucine UUG }	UCU } UCC } Serine UCA } UCG }	UAU } Tyrosine UAC } UAA } Stop codon UAG } Stop codon	UGU } Cysteine UGC } UGA } Stop codon UGG } Tryptophan	U C A G
	C	CUU } CUC } Leucine CUA } CUG }	CCU } CCC } Proline CCA } CCG }	CAU } Histidine CAC } CAA } Glutamine CAG }	CGU } CGC } Arginine CGA } CGG }	U C A G
	A	AUU } AUC } Isoleucine AUA } AUG } Methionine start codon	ACU } ACC } Threonine ACA } ACG }	AAU } Asparagine AAC } AAA } Lysine AAG }	AGU } Serine AGC } AGA } Arginine AGG }	U C A G
	G	GUU } GUC } Valine GUA } GUG }	GCU } GCC } Alanine GCA } GCG }	GAU } Aspartic acid GAC } GAA } Glutamic acid GAG }	GGU } GGC } Glycine GGA } GGG }	U C A G

Triplet Code: $4^3 = 4 \times 4 \times 4 = 64$ codons

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Sixty-one of the 64 codons code for the 20 standard amino acids

- 2. Termination codons:** Three of the codons, UAA, UAG, and UGA, do not code for amino acids but, rather, are termination (also called stop, or nonsense) codons. When one of these codons appears in an mRNA sequence, synthesis of the polypeptide coded for by that mRNA stops.

Characteristics:

- ❑ Usage of the genetic code is remarkably consistent throughout all living organisms.
- ❑ It is assumed that once the standard genetic code evolved in primitive organisms, any mutation (a permanent change in DNA sequence) that altered its meaning would have caused the alteration of most, if not all, protein sequences, resulting in lethality.

Characteristics of the genetic code include the following:

- **1. Specificity:** The genetic code is specific (unambiguous), because a particular codon always codes for the same amino acid.
- **2. Universality:** The genetic code is almost universal, meaning it is nearly the same in all living organisms. This code has remained highly conserved throughout evolution. There are only small differences in how some codons are read. Exception: Mitochondria have a few different codon meanings. For example, in mitochondria, UGA codes for tryptophan (Trp) instead of acting as a stop codon.
- **3. Degeneracy:** The genetic code is degenerate (sometimes called redundant). Although each codon corresponds to a single amino acid, a given amino acid may have more than one triplet coding for it. For example, arginine (Arg) is specified by six different codons. Only Met and Trp have just one coding triplet.

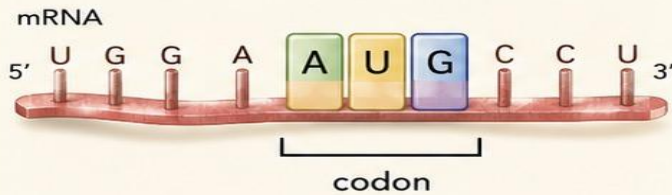
4. Nonoverlapping and commaless: The genetic code is nonoverlapping and commaless, meaning that the code is read from a fixed starting point as a continuous sequence of bases, taken three at a time without any punctuation between codons. For example, AGCUGGAUACAU is read as AGC UGG AUA CAU.

Properties of the Genetic Code

1

triplet

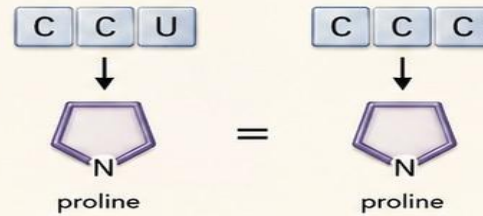
3 RNA nucleotides = 1 codon



2

redundant

Different codons can specify the same amino acid.

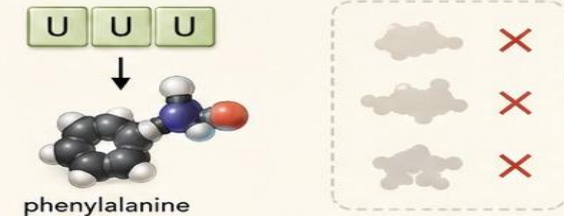


Multiple codons → same amino acid

3

unambiguous

A given codon has only one meaning.



One codon → one amino acid

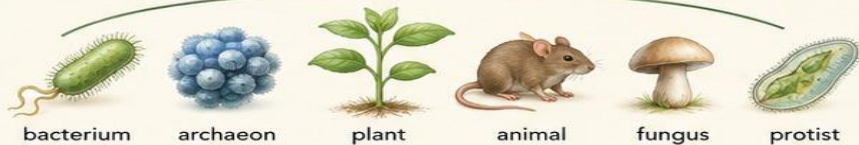
4

nearly universal

The genetic code is nearly the same in all forms of life.



methionine / start

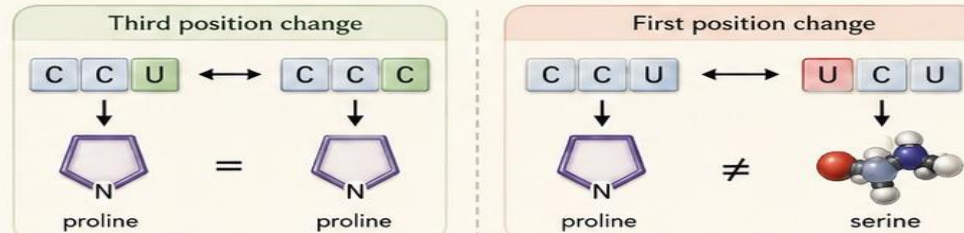


Rare exceptions exist (e.g., in mitochondria and some microbes).

5

conservative

Third-position changes often have smaller effects than first- or second-position changes.



Often same amino acid (smaller effect)

Often different amino acid (larger effect)

Summary

1

triplet

2

redundant

3

unambiguous

4

nearly universal

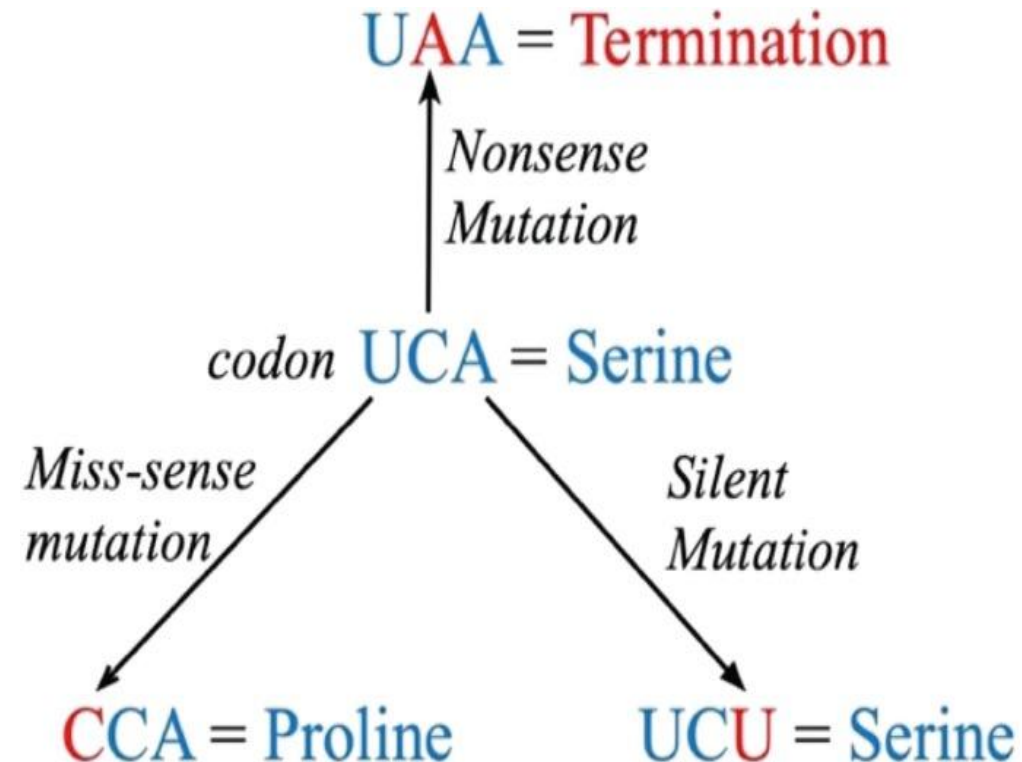
5

conservative

C. CONSEQUENCES OF ALTERING THE NUCLEOTIDE SEQUENCE

- Changing a single nucleotide base (a point mutation) in the coding region of an mRNA can lead to any one of three results
 - 1. Silent Mutation:** The codon containing the changed base may code for the same amino acid. For example, if the serine (**Ser**) codon **UCA** is changed at the third base and becomes **UCU**, it still codes for **Ser**. This is termed a Silent Mutation

- 2. Missense Mutation:** The codon containing the changed base may code for a different amino acid. For example, if the **Ser** codon **UCA** is changed at the first base and becomes **CCA**, it will code for a different amino acid (in this case, **proline [Pro]**). This is termed a Missense Mutation.

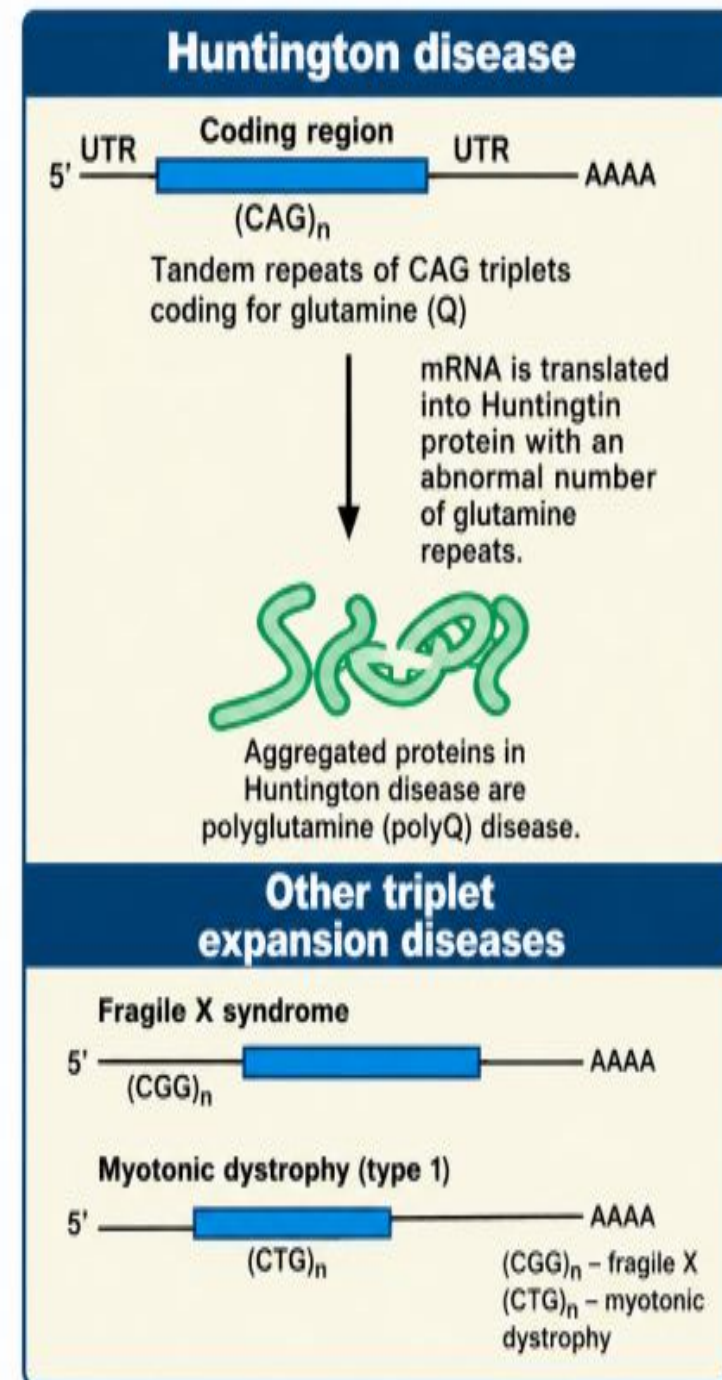


3. Nonsense mutation: The codon containing the changed base may become a termination codon. For example, if the **Ser** codon **UCA** is changed at the second base and becomes **UAA**, the new codon causes premature termination of translation at that point and the production of a shortened (truncated) protein. This is termed a Nonsense Mutation.

4. Other mutations: These can alter the amount or structure of the protein produced by translation.

a. Trinucleotide Repeat Expansion:

- Trinucleotide repeat = a sequence of 3 DNA bases repeated many times.
- Sometimes these repeats become abnormally increased (expanded).
- If expansion occurs in the coding region:
- The extra triplets produce too many copies of one amino acid.
- Example: Huntington disease
- CAG repeat expands → produces excess glutamine (Gln).
- Huntington protein becomes abnormally long.
- The abnormal protein is cleaved into toxic fragments.
- These fragments aggregate in neurons → neurodegeneration



- If expansion occurs in the untranslated region (UTR), it can reduce or stop production of the protein.

- Examples: *Fragile X syndrome
- *Myotonic dystrophy

Fragile X syndrome:

- CAG? No — CGG repeat expansion occurs in the 5' UTR of the FMR1 gene.
- The expanded repeat causes DNA hypermethylation.
- This silences the gene, resulting in very little or no FMRP protein.

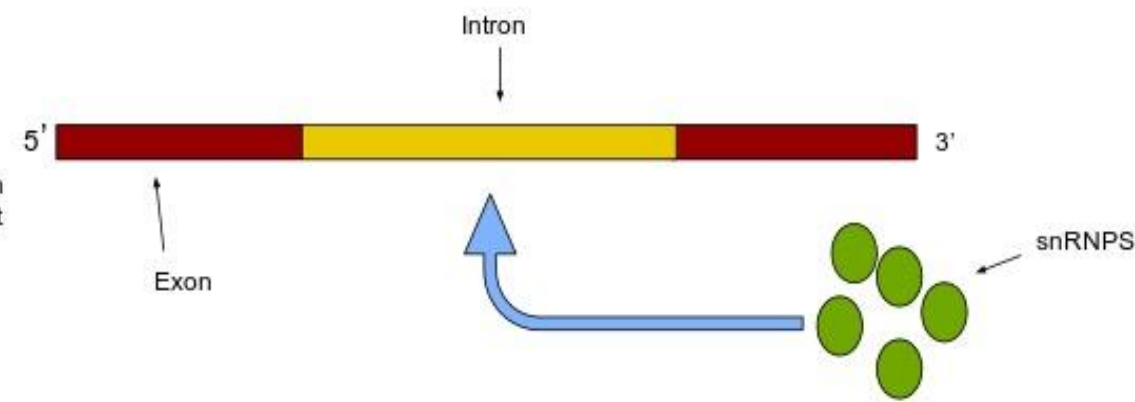
Remember:

Repeat expansion → abnormal protein OR reduced protein → disease

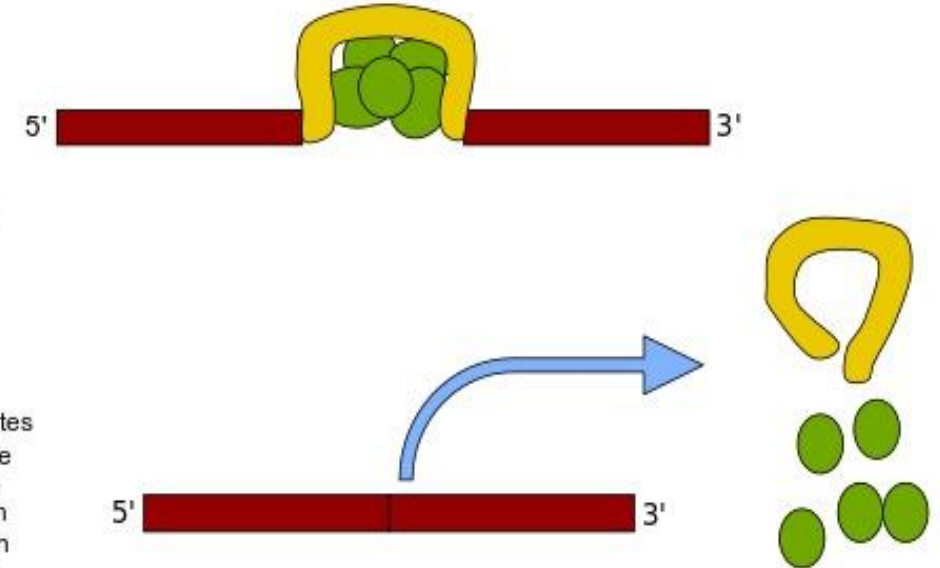
Coding region → extra amino acids → Huntington disease

UTR → decreased gene expression → Fragile X syndrome/ myotonic dystrophy

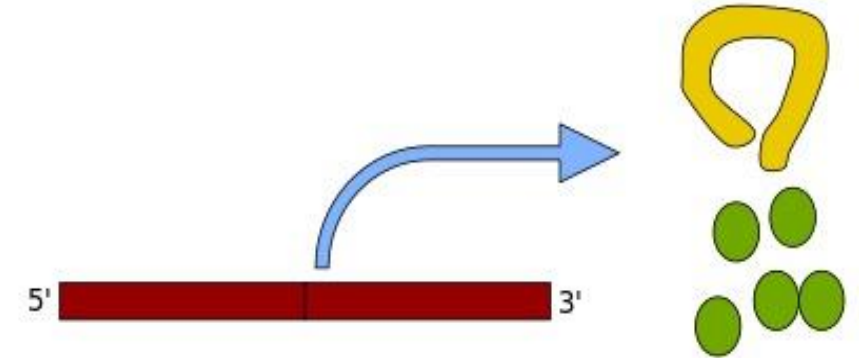
Step 1.
A group of five snRNPS's, or ribonucleoproteins, are needed to bind to the intron of pre-mRNA and remove it to leave only the exons.

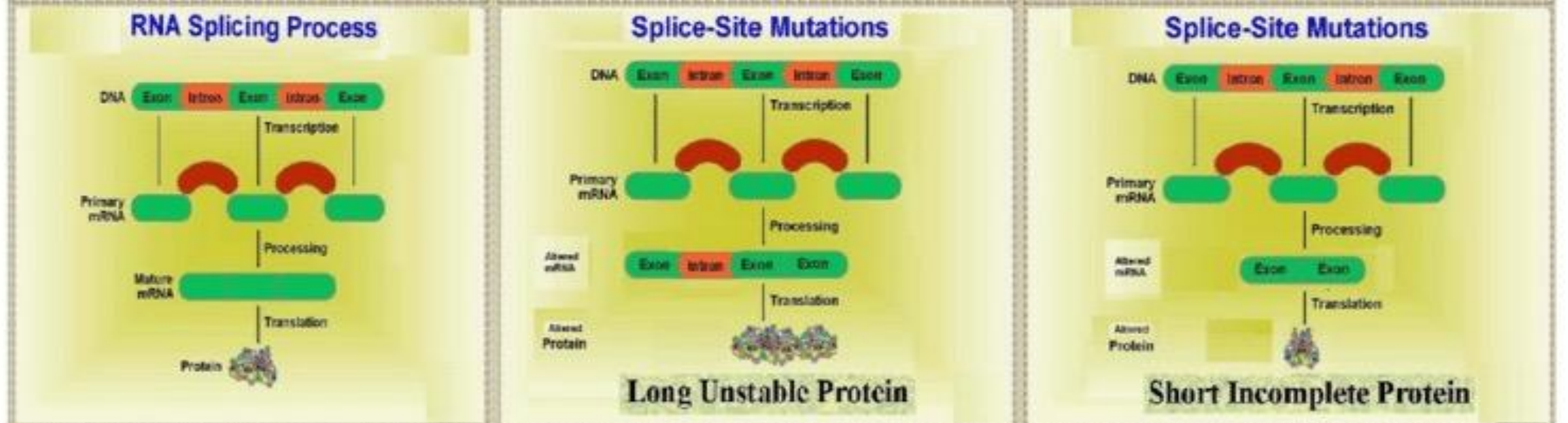


Step 2.
The snRNP's bind to the intron and cause it to fold into bring the 5' and 3' ends of the intron closer together, making a loop. The ends of the exons also move closer together to eventually join together.



Step 3.
The intron detaches and the splice sites connect to make a mature mRNA. The introns were previously thought to be "junk" afterwards but most are used in other processes. The snRNP's detach from the intron and are used for more splicing.





b. Splice Site Mutations: A splice site mutation is a genetic change that happens at the boundary where coding sections (exons) and non-coding sections (introns) meet in a gene's DNA

- Splice sites are specific regions where introns are removed from pre-mRNA.
- A mutation at a splice site can interfere with normal RNA splicing, cause incorrect removal or retention of introns.
- As a result, an abnormal mRNA is produced, abnormal mRNA can lead to production of an abnormal or nonfunctional protein.

Example: Myotonic dystrophy

Triplet repeat expansion causes abnormal RNA splicing. This alters expression of certain genes and contributes to muscle disease.

Remember:

Splice-site mutation → abnormal splicing → abnormal mRNA →
abnormal protein → disease

c. Frameshift Mutations

- Frameshift mutation occurs when 1 or 2 nucleotides are added or deleted from the coding region.
- This changes the reading frame of the mRNA.
- It can produce a completely different amino acid sequence, or a shortened (truncated) protein due to an early stop codon.

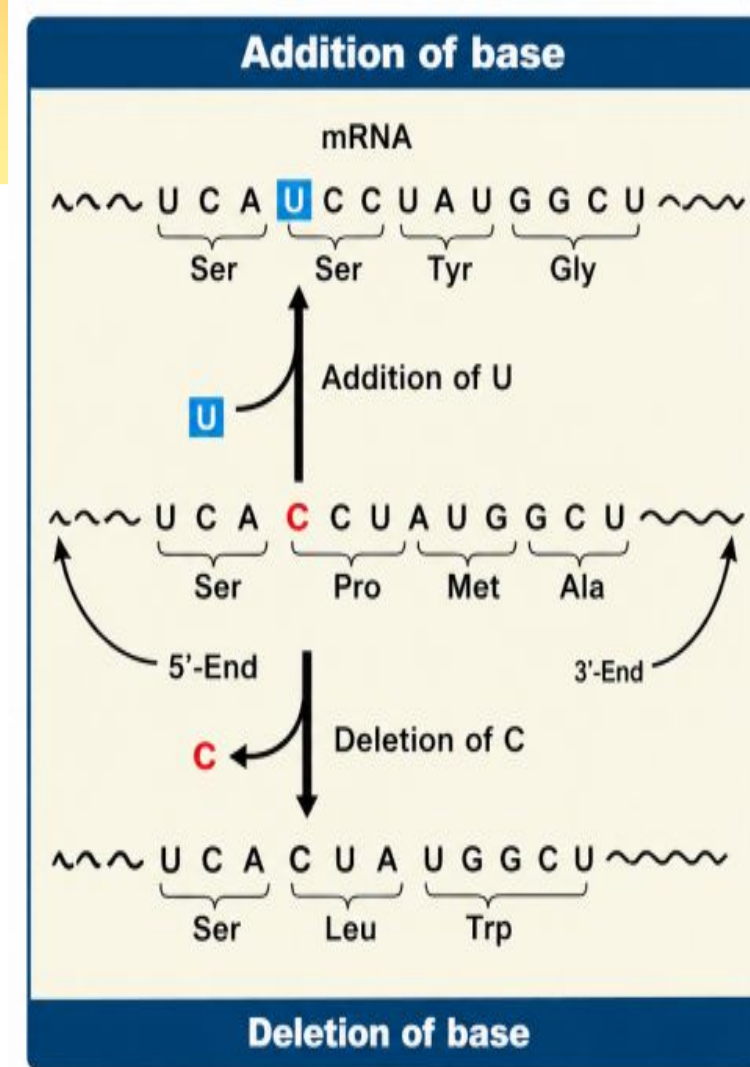
Effect of nucleotide changes

+1 or -1 nucleotide → reading frame changes → usually severe effect.

+2 or -2 nucleotides → reading frame changes → usually severe effect.

+3 nucleotides → one extra amino acid is added; reading frame remains unchanged.

-3 nucleotides → one amino acid is lost; reading frame remains unchanged.



Example: Cystic Fibrosis (CF)

- Most common mutation: $\Delta F508$
- 3 nucleotides are deleted from the CFTR (cf transmembrane conductance regulator) gene.
- This removes phenylalanine (Phe/F) at position 508.
- The reading frame remains

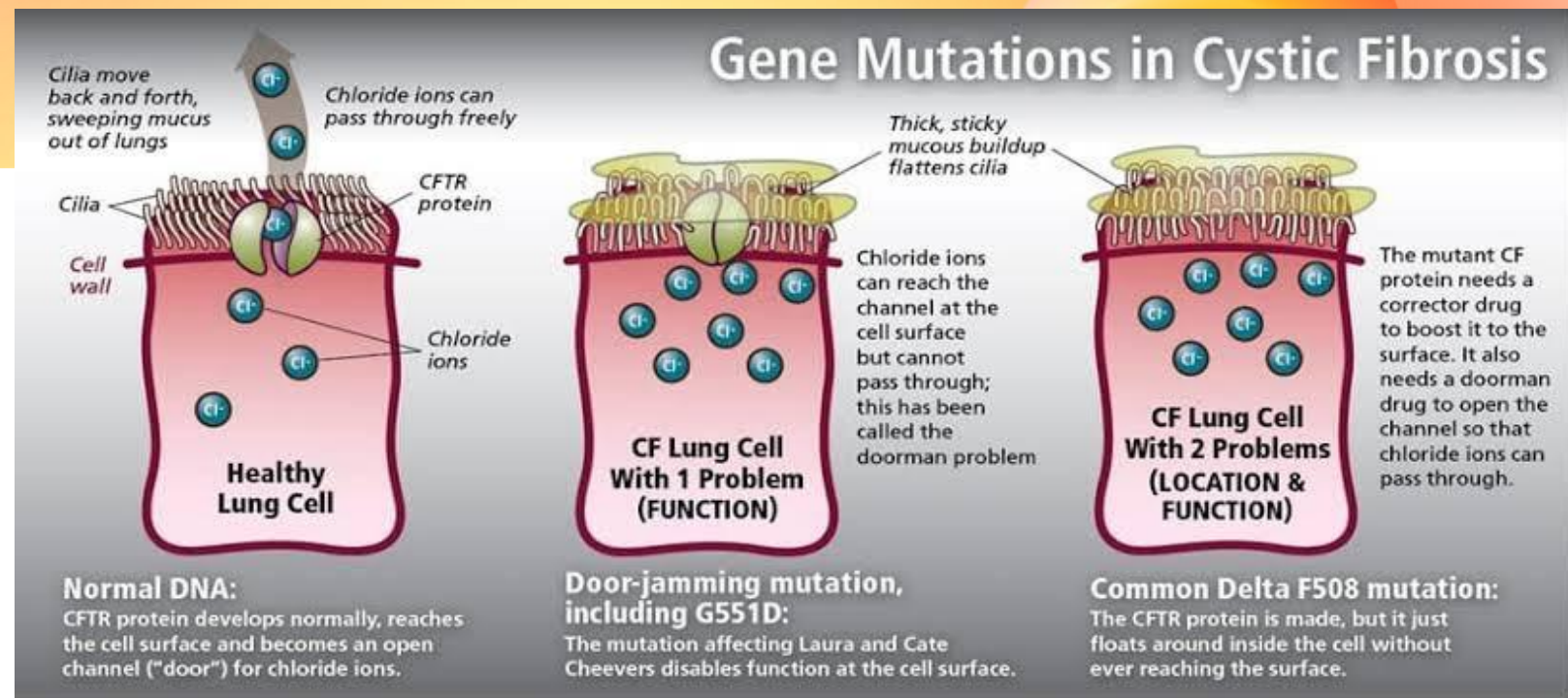
unchanged, but CFTR protein folds incorrectly. Misfolded CFTR is destroyed by the proteasome.

- CFTR normally acts as a chloride channel in epithelial cells.
- Loss of CFTR $\rightarrow$ thick, sticky secretions in the lungs and pancreas.
- This causes lung damage and digestive problems.

Remember:

1–2 nucleotide insertion/deletion $\rightarrow$ Frameshift $\rightarrow$ altered protein

3 nucleotide deletion $\rightarrow$ No frameshift $\rightarrow$ one amino acid lost $\rightarrow \Delta F508$ in CF





THANK YOU